Sleep Monitor

IoT-Based Sleep Monitoring & Therapy System

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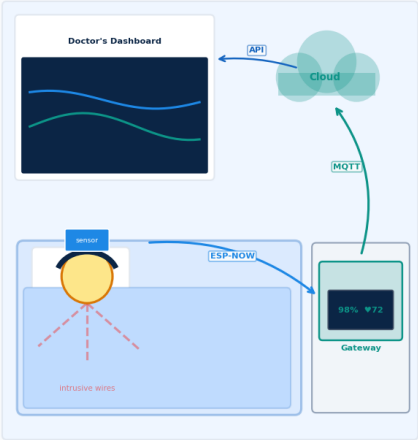
IoT & Mobile Systems — 2025/26

The Problem

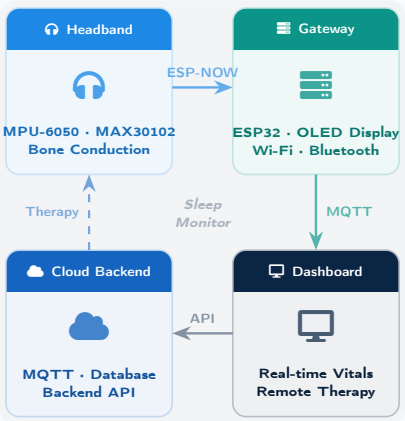
- **Sleep disorders** are prevalent in long-term clinical care, impacting patient recovery and wellbeing
- **Intrusive monitoring** disrupts sleep and provides limited real-time therapeutic insight
- **No remote therapy control** — medical staff must physically intervene at night to adjust treatment



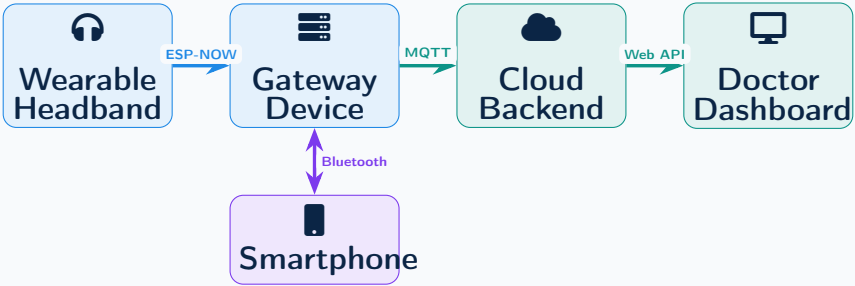
Our Solution

A connected IoT ecosystem that:

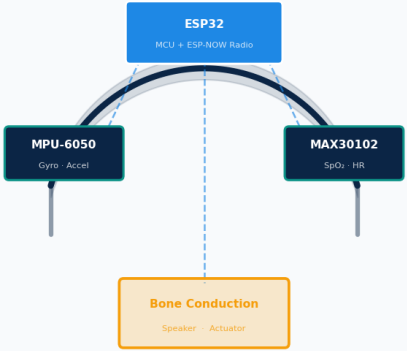
- **Monitors** SpO_2, heart rate & movement in real time
- **Delivers** non-invasive sleep therapy via *bone conduction* sound waves
- **Transmits** data reliably to the cloud via ESP-NOW + MQTT
- **Empowers** clinicians with a live dashboard & remote therapy control



System Architecture



Hardware — Wearable Headband



Sensors

- **MPU-6050**
  3-axis gyroscope + accelerometer
  movement & sleep stage detection
- **MAX30102**
  SpO_2 & heart rate monitoring

Actuator

- **Bone Conduction Speaker**
  Non-invasive therapeutic sound delivery

MCU: ESP32 (ESP-NOW radio)

Hardware — Gateway Device

Role: Bridge between headband and cloud

- **ESP32** — dual role: ESP-NOW receiver + MQTT publisher
- **OLED Display** — real-time vitals readout at bedside
- **Wi-Fi** — relays sensor data to MQTT broker
- **Bluetooth** — one-time Wi-Fi setup via patient's smartphone

Deployable in hospital wards or home-care settings

Communication Protocols

ESP-NOW

Headband → Gateway

- Peer-to-peer, no router
- Ultra-low latency
- IEEE 802.11-based

MQTT

Gateway → Cloud

- Lightweight pub/sub
- Minimal bandwidth
- QoS levels supported

Bluetooth

Phone → Gateway

- One-time Wi-Fi setup
- No extra hub needed
- Easy patient onboarding

Cloud & Backend Stack

MQTT Broker — real-time sensor data ingestion

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Time-Series Database — longitudinal patient storage

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Backend API — data access & therapy command relay

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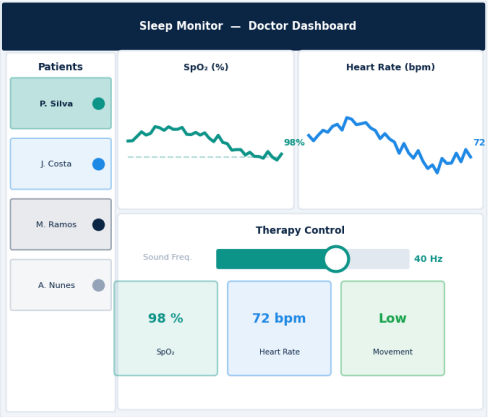
Web Frontend — doctor's real-time dashboard

Doctor's Dashboard



- **Real-time vitals** — SpO_2, heart rate, movement
- **Historical trends** — time-series view per patient
- **Remote therapy control** — adjust bone conduction frequency
- **Multi-patient** management from one interface

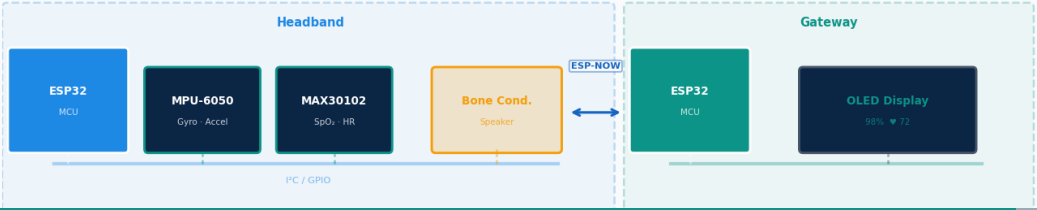
Planned Prototype

Planned Features

- Full sensor integration (MPU-6050, MAX30102)
- Bone conduction actuator
- ESP-NOW, MQTT & Bluetooth links
- Cloud stack + doctor dashboard
- Remote frequency adjustment

Known Constraints

- No integrated battery for demo
- No custom 3D-printed housing yet
- No clinical-scale load testing
- No sleep-stage ML classification



Goals & Expected Impact

- **Non-invasive** monitoring & therapy will reduce patient discomfort
- **Remote control** will eliminate nighttime physical interventions
- **Scalable cloud** architecture designed for multi-ward deployment
- **Bluetooth setup** will lower barrier for home-care adoption

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Thank you — Questions?